Dingkai **Hu**

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Research interests: embodied intelligence and world models, VLM/VLA post-training, agent memory and adaptation, and sim-to-real deployment on real robots.

Education

Tongji University — *Shanghai, China*

B.Eng. in New Energy Materials and Devices (Expected Jun 2029) — *Sep 2025 – Present*

Papers & Manuscripts

A Sparse Lightweight Intent-Aware VLM for Rapid Locomotion Adaptation in Exoskeleton-Assisted Walking
ICRA 2027, under review (submitted Sep 2026) | First author

Built locoVLM on Qwen3.5-2B via structural pruning and vocabulary reduction; maps RGB and IMU streams to a 29-dim decision vector (risk, terrain grade, gait adjustment, per-dim confidence) refined by wearer intent estimated from a single IMU; deployed on Jetson Nano with 78 ms inference latency, reducing execution misalignment from approx. 60% to 13.6%.

ExoMem: From Memory to Matrix Parameters
In preparation (Aug 2026 – Present) | First author

Dual-layer long-term memory for exoskeleton robots: gait cycles are compressed into scene/gait/intent/response block vectors; habitual deviations consolidate into prototypes with online-estimated habit ranges, keeping only rare residual exceptions; fleet-wide priors are expressed as shared matrices and individual residuals as rank-4 factors written in real time via closed-form outer products without gradient passes; compresses the raw stream by approx. 240× at a constant 512B–1KB per-user cost with 0.07–0.16 ms read/write latency.

Research & Competition Experience

Robbyant (Ant Group) Embodied Large-Model Challenge — LingBot-VLA 2.0 — *Online*

Participant — real-robot data collection; LoRA-based VLM training strategy; action-expert alignment — *2026*

- Fine-tuned the open-source LingBot-VLA 2.0 foundation model via LoRA with action-expert alignment; the challenge evaluates models on RoboTwin 2.0 dual-arm simulation tasks.
- Responsible for real-robot demonstration data collection (Franka teleoperation) and for designing the VLM training strategy.

Open-Source Projects

dsh-robotic-harness (30+ stars, MIT)
Embodied-AI research toolkit for DeepSeek Harness: URDF/MJCF asset inspection and conversion, MuJoCo pick-and-place with 6 fault-injection modes, evidence-based diagnostics, and leakage-safe data pipelines with LeRobot export (approx. 110 tools across 14 domains).

mearm-embodied-demo (MIT)
Low-cost (<300 CNY) desktop arm closing the perception–reasoning–action loop: YOLOv8 visual grasping, offline voice (Vosk) and gesture (MediaPipe) control, three-tier LLM intent parsing, and spatial memory for context-aware retrieval.

MoE2Edge (Apache-2.0)
Distills domain-specific edge models from sparse Mixture-of-Experts teachers: routing-based expert selection (DCRS), compact-MoE / dense-student distillation, and INT4 quantization for CPU and consumer-GPU deployment.

Multimodal-Annotation-Studio (MIT)
Schema-driven annotation studio for wearable-robotics multimodal time series (RGB / Depth / FSR / IMU): automatic timeline alignment, gait-phase pre-labeling from FSR, quality tracking, and training-ready CSV/HDF5 export.

Skills

Programming	Python (PyTorch), Git, Linux
Methods	Deep learning; reinforcement learning; diffusion models
Model Training	VLM post-training (LoRA/PEFT, structural pruning, quantization); VLA fine-tuning with action-expert alignment; YOLOv8
Robotics	Real-robot data collection (exoskeleton; Franka arm teleoperation); MuJoCo simulation; edge deployment

Honors & Awards

2026 **National Third Prize**, China Undergraduate Statistical Modeling Competition